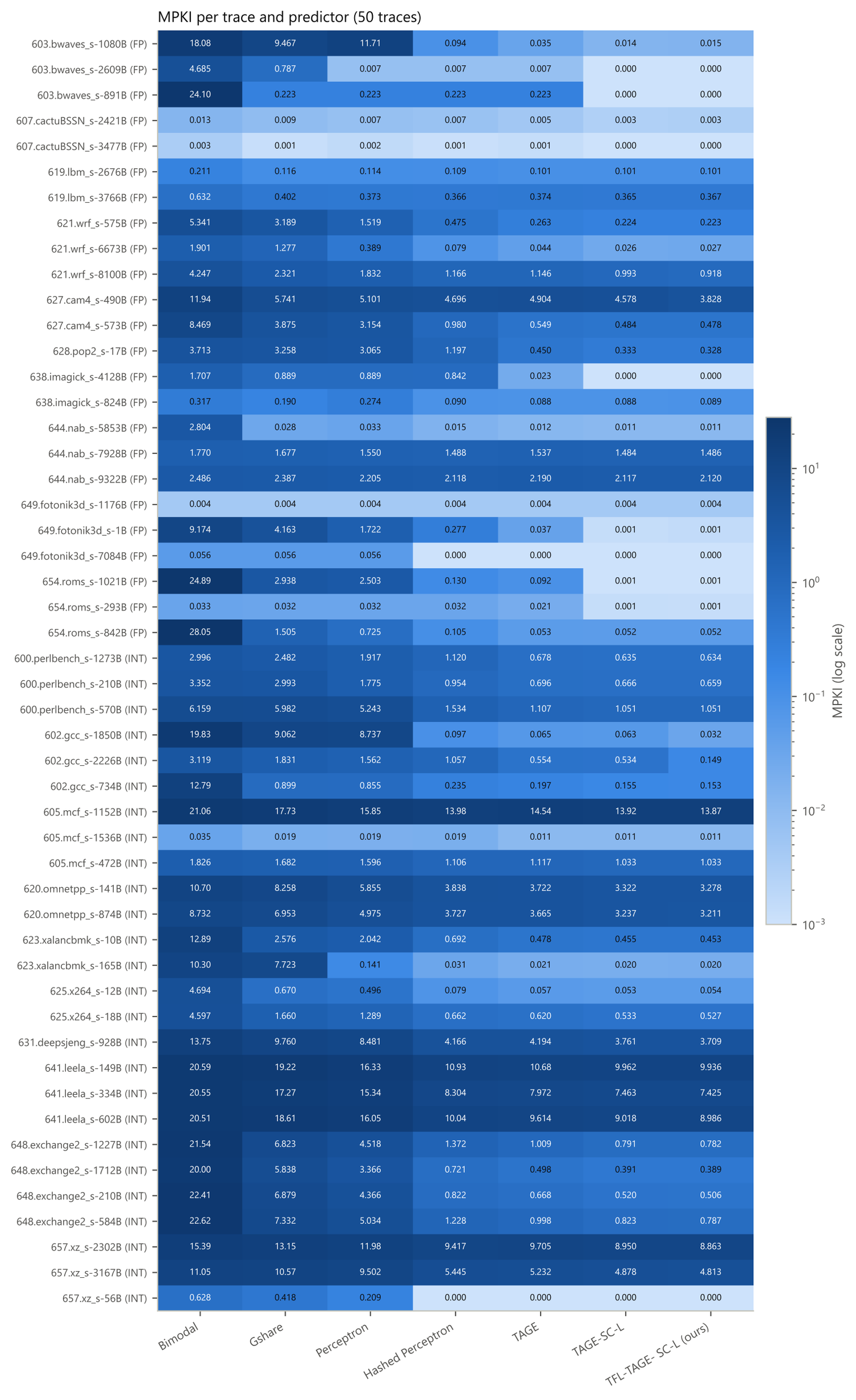

MPKI per trace and predictor (50 traces)
603.bwaves_s-1080B (FP) 18.08 9.467 11.71 0.094 0.035 0.014 0.015
603.bwaves_s-2609B (FP) 4.685 0.787 0.007 0.007 0.007 0.000 0.000
603.bwaves_s-891B (FP) 24.10 0.223 0.223 0.223 0.223 0.000 0.000
607.cactuBSSN_s-2421B (FP) 0.013 0.009 0.007 0.007 0.005 0.003 0.003
607.cactuBSSN_s-3477B (FP) 0.003 0.001 0.002 0.001 0.001 0.000 0.000
619.lbm_s-2676B (FP) 0.211 0.116 0.114 0.109 0.101 0.101 0.101
619.lbm_s-3766B (FP) 0.632 0.402 0.373 0.366 0.374 0.365 0.367
621.wrf_s-575B (FP) 5.341 3.189 1.519 0.475 0.263 0.224 0.223
621.wrf_s-6673B (FP) 1.901 1.277 0.389 0.079 0.044 0.026 0.027
621.wrf_s-8100B (FP) 4.247 2.321 1.832 1.166 1.146 0.993 0.918
627.cam4_s-490B (FP) 11.94 5.741 5.101 4.696 4.904 4.578 3.828
627.cam4_s-573B (FP) 8.469 3.875 3.154 0.980 0.549 0.484 0.478
628.pop2_s-17B (FP) 3.713 3.258 3.065 1.197 0.450 0.333 0.328
638.imagick_s-4128B (FP) 1.707 0.889 0.889 0.842 0.023 0.000 0.000
638.imagick_s-824B (FP) 0.317 0.190 0.274 0.090 0.088 0.088 0.089
644.nab_s-5853B (FP) 2.804 0.028 0.033 0.015 0.012 0.011 0.011
644.nab_s-7928B (FP) 1.770 1.677 1.550 1.488 1.537 1.484 1.486
644.nab_s-9322B (FP) 2.486 2.387 2.205 2.118 2.190 2.117 2.120
649.fotonik3d_s-1176B (FP) 0.004 0.004 0.004 0.004 0.004 0.004 0.004
649.fotonik3d_s-1B (FP) 9.174 4.163 1.722 0.277 0.037 0.001 0.001
649.fotonik3d_s-7084B (FP) 0.056 0.056 0.056 0.000 0.000 0.000 0.000
654.roms_s-1021B (FP) 24.89 2.938 2.503 0.130 0.092 0.001 0.001
654.roms_s-293B (FP) 0.033 0.032 0.032 0.032 0.021 0.001 0.001
654.roms_s-842B (FP) 28.05 1.505 0.725 0.105 0.053 0.052 0.052
600.perlbench_s-1273B (INT) 2.996 2.482 1.917 1.120 0.678 0.635 0.634
600.perlbench_s-210B (INT) 3.352 2.993 1.775 0.954 0.696 0.666 0.659
600.perlbench_s-570B (INT) 6.159 5.982 5.243 1.534 1.107 1.051 1.051
602.gcc_s-1850B (INT) 19.83 9.062 8.737 0.097 0.065 0.063 0.032
602.gcc_s-2226B (INT) 3.119 1.831 1.562 1.057 0.554 0.534 0.149
602.gcc_s-734B (INT) 12.79 0.899 0.855 0.235 0.197 0.155 0.153
605.mcf_s-1152B (INT) 21.06 17.73 15.85 13.98 14.54 13.92 13.87
605.mcf_s-1536B (INT) 0.035 0.019 0.019 0.019 0.011 0.011 0.011
605.mcf_s-472B (INT) 1.826 1.682 1.596 1.106 1.117 1.033 1.033
620.omnetpp_s-141B (INT) 10.70 8.258 5.855 3.838 3.722 3.322 3.278
620.omnetpp_s-874B (INT) 8.732 6.953 4.975 3.727 3.665 3.237 3.211
623.xalancbmk_s-10B (INT) 12.89 2.576 2.042 0.692 0.478 0.455 0.453
623.xalancbmk_s-165B (INT) 10.30 7.723 0.141 0.031 0.021 0.020 0.020
625.x264_s-12B (INT) 4.694 0.670 0.496 0.079 0.057 0.053 0.054
625.x264_s-18B (INT) 4.597 1.660 1.289 0.662 0.620 0.533 0.527
631.deepsjeng_s-928B (INT) 13.75 9.760 8.481 4.166 4.194 3.761 3.709
641.leela_s-149B (INT) 20.59 19.22 16.33 10.93 10.68 9.962 9.936
641.leela_s-334B (INT) 20.55 17.27 15.34 8.304 7.972 7.463 7.425
641.leela_s-602B (INT) 20.51 18.61 16.05 10.04 9.614 9.018 8.986
648.exchange2_s-1227B (INT) 21.54 6.823 4.518 1.372 1.009 0.791 0.782
648.exchange2_s-1712B (INT) 20.00 5.838 3.366 0.721 0.498 0.391 0.389
648.exchange2_s-210B (INT) 22.41 6.879 4.366 0.822 0.668 0.520 0.506
648.exchange2_s-584B (INT) 22.62 7.332 5.034 1.228 0.998 0.823 0.787
657.xz_s-2302B (INT) 15.39 13.15 11.98 9.417 9.705 8.950 8.863
657.xz_s-3167B (INT) 11.05 10.57 9.502 5.445 5.232 4.878 4.813
657.xz_s-56B (INT) 0.628 0.418 0.209 0.000 0.000 0.000 0.000
Bimodal
Gshare
Perceptron
Hashed Perceptron
TAGE
TAGE-SC-L
TFL-TAGE- SC-L (ours)
MPKI (log scale)
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}